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Vocal Individuality in Adult White-naped Mangabey (*Cercocebus lunulatus*) Males at WAPCA's Endangered Primate Breeding Centre, Ghana

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Abstract: Many primate species show vocal individuality. The white-naped mangabey (*Cercocebus lunulatus*) is an Endangered West African primate species, with current feasibility projects aimed at reinforcing wild populations of this species with captive-bred individuals. However, non-invasive monitoring methods to detect translocated individuals are lacking. Passive Acoustic Monitoring (PAM), where self-contained audio devices detect sounds of the target species, can be used to monitor them in the wild. As of today, however, it is not known if individual white-naped mangabeys can be reliably distinguished by their vocalizations. This study aims to answer whether white-naped mangabey males can be differentiated by their voices. We recorded three captive adult mangabey males at the Accra Zoo and extracted vocal features from the resulting spectrograms. Data analysis was carried out using a linear discriminant function analysis and leave-one-out cross validation. The result showed that the adult white-naped mangabey males can be distinguished by their vocalizations with a 59% accuracy. This study, therefore, provides the first evidence of vocal individuality in *C. lunulatus* males. However, further studies are needed to identify translocated mangabey individuals reliably.

Key words: Passive Acoustic Monitoring, vocal individuality, *Cercocebus lunulatus*, acoustic analysis

INTRODUCTION

Vocal individuality is the characteristic of being vocally distinguishable from other individuals and is present in a variety of taxa, e.g., wood owls (*Strix woodfordii*), bottlenose dolphins (*Tursiops truncatus*), blue monkeys (*Cercopithecus mitis stuhlmani*), coppery titi monkeys (*Plecturocebus cupreus*), and baboons (*Papio ursinus*) (Butynski *et al.* 1992; Janik 2000; Delport *et al.* 2002; Rendall 2003; Lau *et al.* 2020). The ability to identify conspecifics is important for a range of behaviors such as mating, resource acquisition, and defense. Many primate species rely on vocal cues for the identification of their group members (Fischer *et al.* 2002; Fukushima *et al.* 2015; Lau *et al.* 2020). The white-naped mangabey (*Cercocebus lunulatus*) is an Endangered primate species endemic to the Upper Guinean rainforest of west Africa (Nolan *et al.* 2019). Habitat loss due to forest fragmentation and degradation and hunting for bushmeat threaten the remaining wild populations of white-naped mangabeys. To reinforce wild populations and hopefully save this species from extinction, individuals can be bred in captivity and released into the wild.

The NGO West African Primate Conservation Action (WAPCA) constructed an Endangered Primate Breeding Centre (EPBC) on the Accra

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Zoo's grounds in Ghana in 2005. In collaboration and coordination with the white-naped mangabey EAZA Ex-situ Program (EEP) the species is bred in captivity to maintain various groups. The primates living in the EPBC participate in this program, which aims to maintain a healthy, genetically viable, and self-sustaining ex-situ population as stock for a translocation program. Four of the mangabeys in the EPBC were confiscated from the pet trade. These originally wild-caught animals add to a higher genetic variability of the reserve population. WAPCA is currently conducting a long-term feasibility study on translocating captive-bred mangabeys into the wild to reinforce wild populations. However, classical methods to monitor released individuals are often invasive as they involve capturing and marking the animals, negatively impacting their health and behavior (Butynski *et al.* 1992; Stowell *et al.* 2019). Passive Acoustic Monitoring (PAM) is a non-invasive monitoring method based on detecting sounds produced by target animals through self-contained audio devices placed in the target animal's habitat. Its biggest advantage is its much wider detection area than camera traps. As seen in the studies by Trolliet *et al.* (2014) and Enari *et al.* (2019) on Japanese macaques, PAM had a higher detectability rate of individuals than camera traps. Other studies have used PAM to estimate occupancy probability, movement patterns, population density and to monitor behavior (Kalan *et al.* 2015, 2016; Wrege *et al.* 2017; Clink & Klinck 2020). The use of vocal features to differentiate between individuals has also been widely established (Butynski *et al.* 1992; Mitani *et al.* 1996; Rendall 2003; Price *et al.* 2009; Budka *et al.* 2015; Clink & Klinck 2020). The production of sound starts with the lungs providing an airstream. The larynx, by opening or closing the laryngeal valve controlled by the vocal folds, modulates the airstream, converting it into sound in a series of puffs (Fitch 2002). Longer and thicker vocal folds can create lower pitches, and therefore a lower fundamental frequency. For this reason, the larynx can be described as the "source" of the sound. The created source signal then radiates through the supralaryngeal vocal tract (i.e., the airways of the pharynx, mouth, nasal cavities, lips, and nostrils).

Primate alarm calls can be heard over long distances and are emitted frequently, making them easier to record in the wild than other calls (Butynski *et al.* 1992; Mitani *et al.* 1996). Little is known about the white-naped mangabey's vocal repertoire, but studies on its close relatives, the sooty mangabey (*Cercocebus atys*) and the red-capped mangabey (*Cercocebus torquatus*), reveal similarities in alarm calls (Range & Fischer 2004; Bouchet *et al.* 2012). Both male and female red-capped mangabeys emit a "wahoo" call that consists of two syllables the "wa" and the "hoo". During "wahoo" calls, males often carry their tails arched over their back, gape, yawn and shake. These signs of physical agitation are associated with the "wahoo" call, which is associated with a high level of alertness in red-capped mangabeys. Similarly, baboons (*Papio ursinus*) have a "wahoo" call, which, depending on the situation, can either function as contact, alarm, or contest calls (Fischer *et al.* 2002). Therefore, it can be presumed that the white-naped mangabey's "wahoo" call functions primarily as an alarm call because of the close relationship of *C. lunulatus* to *C. torquatus* (Bouchet *et al.* 2012).

This study aims to answer whether white-naped mangabey adult males can be distinguished by their "wahoo" calls. Recognizing the individuality in their calls will allow us to inform monitoring techniques for the released primates further and assess the translocation success. We chose the "wahoo" call specifically because it is a loud and frequently emitted call of the mangabey and can, therefore, be easily recorded in the field (Range & Fischer 2004). Our hypothesis is that individual males can be reliably distinguished by their vocalizations and vocal cues because many other primate species show vocal individuality. This would introduce a new non-invasive monitoring method for this species, which would be especially useful after releasing captive-bred mangabeys into the wild.

METHODOLOGY

Study Site and Subjects

Data were collected at the EPBC at Accra Zoo in Accra, Ghana (Figure 1). Accra is the capital city of Ghana and is located at the coast with two wet seasons (March-July, September-November) where the average temperature ranges from 22˚C to 25˚C and from 25˚C to 28˚C in the dry seasons (Kayaga *et al.* 2021).

Accra Zoo houses four white-naped mangabey groups. For this study only three were considered, since one enclosure did not house any adult males (see Table 1). From here on, the first group is designated as the "Ape's group" and consists of the adult alpha male Ape and two other adult males, Klaus and Kwame. Ape is a former pet that was confiscated and rehabilitated at the EPBC. The female Accra and the two juvenile males, Yellow and Austin, live in this group. Klaus and Kwame were not considered in

Figure 1. Location of Accra Zoo, where the Endangered Primate Breeding Centre is located, within the Achimota Forest in the city of Accra, Ghana.

this analysis since, as non-dominant males, they did not emit calls frequently enough. The second group, from here on called "Yaw's group", consists of the adult male Yaw, the adult female Ivy, the subadult female Mama and the juvenile female Esi. Lastly, the third group, from here on named "Kweku's group", consists of the adult male Kweku the two adult females Ziggy and Chaachele, the juvenile female Mercedes and the two juvenile males Tommaso and Peter. The enclosure of Ape's group is 100 m², Yaw's group 108 m² and Kweku's group is 131 m². All groups are fed with fruits and vegetables twice a day. Water is available ad libitum. The individuals of each group can hear and see each other. Additionally, the zoo allows visitors from 9 am to 5 pm daily.

Data Collection

The "wahoo" calls of the adult males of each group were recorded using a H4next Handy Recorder and a windshield. The recorder was positioned on the ground, outside the keeper's entrance of each enclosure. The distance from the calling individual and the recorder ranged from 2 to 17 m. Data were collected from May to June 2022, between 6am to 9am Monday to Friday. The adult males of each group were recorded opportunistically before visitors arrived at the zoo. Each call was followed by a note of the individual's name, location in the enclosure, and behavior shown during the call. The ethogram by Abelló *et al.* (2018) was used to identify the individuals' behaviors, even though it did not play a role in the analysis.

Analysis

The fourteen audio recordings of the "wahoo" call with the least background noise were selected from each alpha male (Ape, Yaw, Kweku). The call was divided into two syllables "wa" and "hoo". Praat version 6.3.23 (Boersma & Weenink 2023) was used to extract the following acoustic features: duration, dominant frequency, and fundamental frequency of each syllable of the "wahoo" call, and the first to fourth formants of the full call (N = 10 acoustic features). Additionally, spectrograms were drawn in Praat to extract features visually with the following settings: view range (Hz) 50.0 to 3000.0, window length 0.008 and dynamic range 50.0. Advance settings were left with the standard values. A representative spectrogram is presented in Figure 2. The call duration was extracted visually from the spectrogram. The dominant frequency was determined by selecting the syllable in the

Table 1. White-naped mangabey (*C. lunulatus*) groups in the Endangered Primate Breeding Centre at Accra Zoo included in this study, with their dominant males highlighted in bold letters. See McGraw *et al.* 2011 for age categories.

	Adult Males (> 7 or 8 years)	**Adult Females**	**Juvenile Males (1 to 6 years)**	**Juvenile Females (1 to 4.5 years)**
Group 1: Ape's group	**Ape (former pet)**, Kwame, Klaus	Accra	Yellow, Austin	/
Group 2: Yaw's group	**Yaw (captive bred)**	Ivy	/	Mama, Esi
Group 3: Kweku's group	**Kweku (captive bred)**	Ziggy, Chaachele	Tommaso, Peter	Mercedes

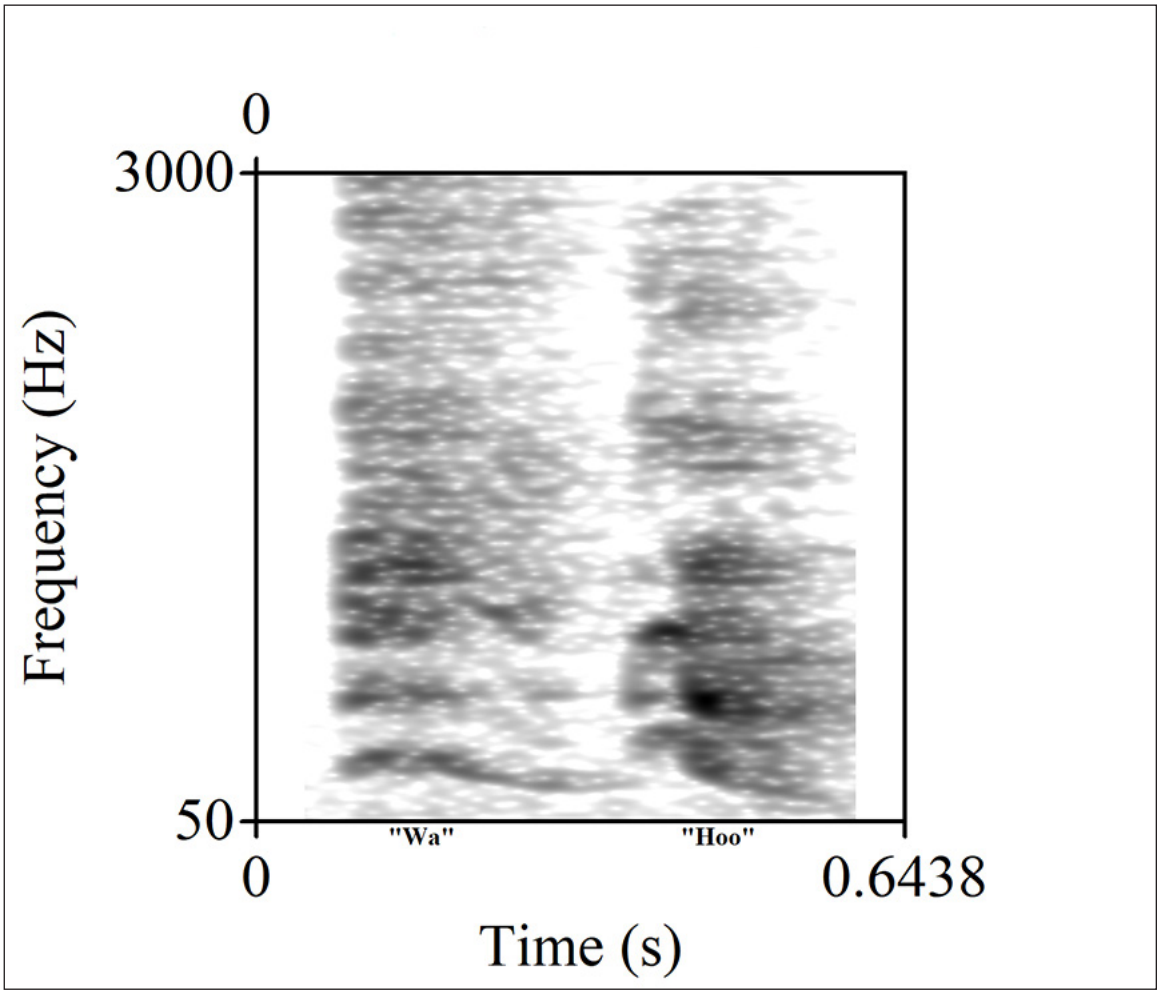

Figure 2. Representative spectrogram of a white-naped mangabey (*C. lunulatus*) male (Yaw) "wahoo" alarm call at Accra Zoo.

spectrogram and then viewing the spectral slice with dominant frequency automatically appearing on the top axis. The fundamental frequency was determined using Praat's pitch tracker in the standard settings, and the four formants were identified using spectrum analysis to formants. It should be noted that whilst we used Praat's formant tracker to identify the first four formants of each "wahoo" recording, we did not test if these bands result from resonances in the vocal tract, or if they are independent from the fundamental frequency and its harmonics (Price *et al.* 2009). The recorded "wahoo" calls had a higher fundamental frequency than human voices (around 200-300 Hz), which might have affected Praat's formant tracker in extracting formant frequencies. Previous studies have conducted similar analyses using Praat for non-human primate recordings and yielded reliable results (Price *et al.* 2009; Gamba *et al.* 2012; Grawunder *et al.* 2022).

We tested all features for normal distribution to find out significant differences in these acoustic features between individuals. We also conducted a one-way ANOVA for the normally distributed features and the Kruskal-Wallis test for non-normally distributed features with the significance level set at 0.05 (Butynski *et al.* 1992; Price *et al.* 2009). We conducted a linear discriminant function analysis (DFA) to identify differences in the vocalizations between individuals (Fischer *et al.* 2002). DFA is a commonly used data classification method that maximizes the ratio of the between-class variance to the within-class variance, guaranteeing maximum class separability (Tharwat *et al.* 2017). By using input features DFA estimates the maximum difference between calls from each individual and provides a classification procedure that assigns each call to its appropriate individual (correct assignment) or another individual (incorrect assignment) (Fischer

et al. 2002; Lau *et al*. 2020). Additionally, to identify the model's accuracy, we conducted a leave-one out cross validation where each unit is classified by the functions derived from all units other than one (Bouchet *et al.* 2012, Price *et al.* 2009).

RESULTS

We found significant differences in seven acoustic features of the "wahoo" call between the white-naped mangabey (*C. lunulatus*) males at Accra Zoo, presented in Table 2. Differences were found in the dominant frequency of the "hoo" ($p = 0.0048$), the fundamental frequency of the "wa" ($p = 0.0154$), formant 2 ($p = 0.0128$), formant 3 ($p = 0.0004$), formant 4 ($p = 0.0056$), duration of the "wa" ($p = 0.0011$), and duration of the "hoo" ($p = 0.0019$).

The linear discriminant function analysis yielded an average correct assignment of 59% and the leave one out cross validation validated it with 76% accuracy. The scatter plot of the two linear discriminant functions to illustrate the clustering of "wahoo" alarm calls in relation to caller identity is presented in Figure 3.

DISCUSSION

This study provided the first evidence of vocal individuality in white-naped mangabey (*C. lunulatus*) males. We successfully identified males based on multiple acoustic features of their "wahoo" calls using linear discriminant function analysis. The leave-one-out cross validation revealed that the DFA was accurate when this data set was used.

Vocal individuality occurs because of differences in the morphology of the sound-producing organs, e.g., lungs, larynx, nose, and mouth, between individuals (Fitch & Hauser 1995; Lau *et al*. 2020). The vocal tract acts as a filter by removing certain frequencies from the source signal. The size of an individual changes the size of the vocal tract, leading to larger individuals having lower fundamental frequencies while formant dispersion decreases with size (Fischer *et al.* 2002).

Larger lung capacities allow longer calls, increasing the duration of the "wa" in the "wahoo" calls of baboons. Furthermore, macaques can tell an individual's body size by the formants of its call (Fukushima *et al*. 2015).

We successfully classified 59% of the recorded calls in the test data set. Compared to similar studies, this probability rate is lower (blue monkeys: 100% [Butynski *et al.* 1992], baboons: 71.2% [Fischer et al. 2002] and 65% [Rendall 2003], coppery titi monkeys: 83% [Lau *et al.* 2020]). We believe this is because we analyzed the "wahoo" call of white-naped mangabey males that, similar to the "wahoo" call of baboons, may have multiple functions (i.e., alarm and contest call, Fischer *et al.* 2002). Bouchet *et al.* 2012 found that individual distinctiveness varied between the call types in the vocal repertoire of red-capped mangabeys. They found the highest level of individual distinctiveness in threat and contact calls and lower levels of individuality in loud or alarm calls. This is supported by Price *et al.* 2009 who found that the multifunctional "pyow" call (alarm and contact call) of putty-nosed monkeys can be distinguished between individuals while the "hack" alarm call cannot. The social function hypothesis aims to explain these observations by stating that there is a positive correlation between a call's relevance during social interactions and its morphological complexity (Bouchet *et al*. 2012). Individual discrimination is important during intragroup social interactions, so calls used for social purposes are more individually distinctive than calls used in other situations. However, identifying the alarm calling individual can still be useful to respond in accordance with the caller's reliability. This might explain why vocal individuality is still present in alarm calls (Price *et al.* 2009).

Our low probability rate could be attributed to our small data set. According to the leave-one-out cross validation, the data set was large enough to prevent overfitting; however, a larger training data set could increase the probability of correctly assigned recordings. Another reason for our low probability rate could be that a large portion of our acoustic features were formants analyzed using PRAAT's formant tracker using default settings. Although previous studies have achieved significant results with this methodology, PRAAT's formant tracker—when using default parameters—is optimized for human voices and may not be well-suited for white-naped mangabey vocalizations (Price *et al.* 2009). Previous studies have found that tracker results were biased considerably by the sex of the human speaker (Schiel & Zitzelberger 2018). Therefore, more acoustic features would likely have increased our probability, and future studies should consider adjusting the formant analyzer to the research subject's vocal tract (Chapman & Weary 1990; Lau *et al*. 2020).

Because white-naped mangabey males show vocal individuality in their "wahoo" calls, we believe that PAM can be a great addition to previous monitoring methods for this species. As DFA needs a training data set to identify individuals, as many

Table 2. Means of the acoustic features of the male white-naped mangabey "wahoo" call per individual and the p-values calculated with one-way ANOVA and Kruskal-Wallis test. Statistically significant features are highlighted in bold letters. aNon-normally distributed data calculated with the Kruskal-Wallis test.

Acoustic Feature	Individual	Mean	p-value
Dominant Frequency "Wa" in Hz	Ape	675.86	0.0964a
	Kweku	775.64	
	Yaw	635.43	
Dominant Frequency "Hoo" in Hz	Ape	269.07	0.0048
	Kweku	252.14	
	Yaw	284.50	
Fundamental Frequency "Wa" in Hz	Ape	269.07	0.0154a
	Kweku	252.14	
	Yaw	284.50	
Fundamental Frequency "Hoo" in Hz	Ape	198.21	0.7070
	Kweku	204.07	
	Yaw	205.86	
Formant 1 in Hz	Ape	1040.37	0.1630
	Kweku	1127.86	
	Yaw	1089.79	
Formant 2 in Hz	Ape	1892.38	0.0128
	Kweku	2004.20	
	Yaw	1981.37	
Formant 3 in Hz	Ape	2900.00	0.0004
	Kweku	3068.57	
	Yaw	2934.59	
Formant 4 in Hz	Ape	3973.96	0.0056
	Kweku	4097.80	
	Yaw	3940.12	
Duration "Wa" in seconds	Ape	0.26	0.0011
	Kweku	0.24	
	Yaw	0.28	
Duration "Hoo" in seconds	Ape	0.19	0.0019
	Kweku	0.24	
	Yaw	0.25	

individuals as possible (ideally all) would need to be recorded and identified in a habitat or wild population to reliably identify the released ones (Budka *et al.* 2015).

With the ongoing advancement of artificial intelligence, the question arises whether this technology can be utilized for future monitoring methods (Ang *et al.* 2023). Bird recognition apps that use artificial intelligence to identify bird species by their voices have already been developed (Tian & Sun 2022, Ware *et al.* 2023). Similar apps for identifying monkey species are conceivable. However, there are currently no apps that offer acoustic detection at the individual level rather at

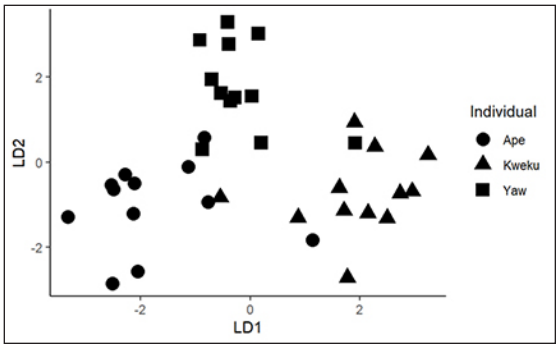

Figure 3. Scatter plot using the two linear discriminant functions of the "wahoo" calls of three adult male white-naped mangabeys showing scores from 10 acoustic features, overlaid with caller identity. LD1 and LD2 are the two axes created by the linear discriminant function analysis.

a species level. Furthermore, actively recording spontaneous calls of white-naped mangabeys with a smartphone instead of a PAM system seems difficult. The possibility of encountering a pre-recorded individual in the field is very slim.

Future studies should analyze the white-naped mangabey's vocal repertoire to understand better this species' contact, loud, and alarm calls. According to Range & Fischer (2004), sooty mangabeys' social calls include grunts, twitters, screams, growls, and other threat and copulation calls. It can be assumed that this is similar for members of the same taxon - Cercocebus, and our study should be repeated using social calls to see if this increases the probability rate of the DFA. However, social calls are difficult to record in the wild since often they cannot be heard over long distances. Furthermore, future studies should investigate the exact purposes of the "wahoo" call because of its similarity to the baboon's "wahoo" call, which can be used in different contexts (Fischer *et al.* 2002). Additionally, further research is needed to determine if an individual's acoustic features remain constant over long periods of time, over individual growth, and after changes in group composition to ensure reliable monitoring (Butynski *et al.* 1992; Price *et al.* 2009). Lastly, future studies should also investigate vocal individuality in white-naped mangabey females.

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